

Carter Delavan-Hoover

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ENGINEERING EXPERIENCE

TSMC Arizona | Phoenix, AZ - Nanolithography Equipment Engineer Intern

Jun 2026 – Aug 2026

- Independently developed and validated a pump reuse qualification protocol from the ground up by designing and testing flushing procedures, collecting performance data, and **successfully qualifying 3 pumps for reuse** through multi-week testing.
- Established the project's long-term experimental framework to evaluate how photoresist properties affect pump cleanability and reuse, enabling future engineers to continue testing and supporting a project with **~\$1M/year potential savings**.
- Led an independent effort to **reduce pump pre-qualification time by 50%**, from 2 weeks to 1 week, by analyzing fab workflows and eliminating operator downtime caused by chemical replenishment and recipe-transition delays.
- Spent extensive time in a semiconductor cleanroom overseeing and participating in pump and **fab-tool repairs**, working directly with operators and maintenance personnel to troubleshoot equipment and improve production workflows.

Terrapin Works | College Park, MD - Scanning Technical Coordinator

Sept 2024 – Present

- Performed ~0.1 mm-accurate 3D metrology using Romer Absolute Arm, Artec Leo, and Artec Spider on mechanical parts, consumer products, artwork, and full-body scans.
- Processed scans in Artec Studio, Geomagic, SolidWorks, and Blender to create manufacturable CAD models for prototyping, analysis, and reverse engineering.
- Diagnosed FDM printer mechanical/calibration issues and authored a repair guide to improve system uptime.

NASA L'SPACE Program | College Park, MD- Lead Project Engineer

Sept 2024 – Dec 2024

- Led systems engineering for a dual-rover lunar mission, driving architecture, requirements, mass allocation, and subsystem interfaces while coordinating a 5-person multidisciplinary team.
- Conducted **~20 engineering trade studies** spanning locomotion, chassis, power, and thermal systems; developed CONOPS, N² matrices, verification/risk matrices, and PDR documentation to NASA standards.
- Developed Siemens NX models for chassis, scientific payloads, solar array, and subsystem packaging to support integration and mass tracking.

UMD Loop | College Park, MD - Suspension Team Member

Sept 2024 – May 2026

- Designed a two-linkage suspension with independently moving arms and a custom foam dampener replacing traditional springs.
- Performed load, stress, deflection, and buckling analyses; developed MATLAB tools to optimize dampener geometry and material properties under packaging and load constraints.
- Created full suspension CAD in SolidWorks, including custom sheet-metal and COTS components; resolved chassis/steering integration issues to ensure proper load paths and articulation.

EDUCATION

University of Maryland, College Park

Exp. Graduation May 2027

Bachelor of Science in Mechanical Engineering, Minor in Astronomy

TECHNICAL SKILLS & ADDITIONAL

CAD/CAE: SolidWorks, Siemens NX, Fusion 360, Geomagic, Artec Studio, FEA

Analysis: MATLAB, Statics & Dynamics, FEA, Requirements & Risk Analysis

Manufacturing: DFM, Metal Machining, FDM 3D Printing, 3D Metrology

Programming: Python, Matlab, Git/GitHub, Excel VBA

Certifications: SolidWorks CSWA • Siemens NX Certification